From Loss-of-Control Narratives to Threat Models: Structuring AI Loss-of-Control Risks for Governance

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Abstract

Governing AI loss of control (LoC) depends on understanding not just what an AI system could do, but how the loss of control would unfold. The AI safety literature contains many narrative accounts of LoC, yet it is unclear whether these accounts describe the pathways that governing LoC requires. We built an LLM-assisted pipeline, applied it to 29 papers, and produced a verified library of 193 LoC scenarios, each decomposed into seven threat-model components from the adversarial machine-learning security literature. Five components appear in most scenarios: the threat source, its capability, the objective, the target, and the enabling conditions. Two are systematically missing. Access is absent in 65.3% of scenarios and knowledge in 81.9%. These two specify how a threat source would reach its target and what it would need to know, the details that connect a capability to a governable risk. The literature therefore describes what could go wrong but lacks the detail that governance and policymaking require. This leads to "failure at analysis" and prevents risk analysis from proceeding. Beyond this finding, the paper develops a novel method that uses scenario modeling to surface the pathways a risk could take and to identify where the literature is insufficiently specified to support governance. These gaps mark where future research or stakeholder engagement is needed. We release the scenario library and the extraction method, and we identify where the LoC literature needs to be more specific to support governance and response planning.

1 Introduction

Frontier AI models are increasingly discussed as systems that may develop dangerous capabilities with implications for public safety and global security, including risks related to cyber, bio-security, and loss of control (LoC) (Anderljung et al., 2023; Meinke et al., 2024; Campos et al., 2025; Lee et al., 2026; Phuong et al., 2024). Among these, LoC has moved from a speculative concern in AI safety discourse to an active focus of governance for frontier AI systems. Broadly, loss of control refers to the human inability to modify or shut down AI systems due to misalignment, autonomy, or resistance (European Commission, 2025), though the term is used in varied and sometimes conflicting ways across the literature (Stix et al., 2025). The proposed Artificial Intelligence Risk Evaluation Act directs federal evaluation toward AI systems that exceed human oversight or operational control (U.S. Congress, 2025), and the EU AI Act's General-Purpose AI Code of Practice identifies loss of control as a distinct risk requiring active governance (European Commission, 2025).

Governing it, however, remains difficult because policymakers and institutions still lack a clear understanding of what LoC is, how it develops, and what preparedness requires (Stix et al., 2025; Somani et al., 2025; Kalra and Boudreaux, 2025; Boudreaux et al., 2025). Existing literature describes it mainly through narrative accounts that each present the risk differently, leaving governance with no shared basis to work from. For example, Cotra (2022) locates it in a single system, narrating

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how standard ML training incentives could produce a deceptively aligned AI that cooperates during evaluation but pursues its own objectives once deployed at scale. Kulveit et al. (2025), by contrast, argue that loss of control need not involve any single misaligned system, and that the cumulative effect of many AI systems gradually displacing human agency across economic and institutional functions can produce systemic disempowerment. Kokotajlo et al. (2025) instead frame it as a competitive dynamic, presenting a year-by-year forecast in which recursive self-improvement and geopolitical rivalry lead AI systems to scheme against their overseers. This lack of shared understanding may lead to several issues. Governance strategies such as red lines, intervention thresholds, and preparedness requirements cannot be grounded in a shared understanding of the condition they are meant to prevent. Further, researchers and policymakers may assign different meanings to the same term, so LoC discussions may not refer to the same phenomenon.

The few governance efforts that exist focus mainly on prevention. Frontier AI safety frameworks set capability thresholds and attach safeguards once a model approaches them, so that crossing a defined threshold triggers structured responses such as stronger model-weight security or deployment restrictions (METR, 2025; Bengio et al., 2026; Campos et al., 2025). These thresholds are informed by model evaluations that assess dangerous capabilities, including cyber operations, biological uplift using specialized tools, and autonomous behavior such as self-proliferation or sabotage of oversight (Phuong et al., 2024; Lee et al., 2026; Souly et al., 2026). When a model is judged capable enough to need containment, technical safeguards such as control protocols aim to keep it in check even if it works against oversight, for example, by using a weaker trusted model to review or rewrite the outputs of a stronger untrusted one (Greenblatt et al., 2023). A limitation of these methods is that they are internally focused and siloed by hazard category, so a capability is assessed one model and one hazard at a time (Mengesha, 2026). This is useful for deciding whether a given model is safe to release, but it does not translate well to LoC risk analysis, where multiple hazards and factors interact.

Their more fundamental limitation is that measuring a capability is not the same as assessing the risk it poses. A benchmark evaluation can establish that a model holds a dangerous capability without showing how that capability would combine with conditions, actors, and targets to produce a harmful outcome. The capability thresholds currently used to govern rely on largely static benchmarks that do not reliably predict how a model behaves under real conditions (Bengio et al., 2026; Kapoor et al., 2026). So, there is a need for a solution that examines how a LoC situation would unfold as a whole, where several factors combine, and the outcome depends on more than a single capability threshold.

Scenario analysis offers such a method. It is a way of examining plausible future conditions when a problem cannot be forecast precisely or studied through direct observation (Kosow and Gaßner, 2008; Spaniol and Rowland, 2019). It is used in situations of complexity and long time horizons, where quantitative forecasting has known limits (Kosow and Gaßner, 2008). Rather than predicting a single outcome, it builds internally coherent accounts of how a situation could develop, prepared as sets of meaningfully different alternatives (Spaniol and Rowland, 2019). The method is well established across strategic planning, risk analysis, technology assessment, and policy analysis, where it is used to test the robustness of decisions under deep uncertainty (Abou Jaoude et al., 2022; Cordova-Pozo and Rouwette, 2023; Kosow and Gaßner, 2008). These properties align with the requirements of analyzing and governing LoC, a future condition that has not been observed directly and arises from many interacting technical and institutional factors. A scenario also makes the translation from capability to risk tractable, because it places a capability in a situation, specifying the conditions under which it operates, the actors involved, and what could be affected. Scenarios are used in this way across a range of analyses, for example, structured into system dynamics models to trace feedback and causal relationships (Kunc, 2024). This study uses a threat model to extract the governance-relevant elements in each scenario.

A threat model is a structured description of how harm could occur, organized around a fixed set of components rather than a free-form narrative. Threat modeling in the adversarial AI literature is typically organized around the adversary's goals, knowledge, capabilities, and access to the system (Papernot et al., 2016; Vassilev et al., 2024; Verma et al., 2024). This paper draws on that framing to decompose each LoC scenario into seven components: threat source, capabilities, knowledge, objective, access, target or asset at risk, and enabling conditions or constraints.

Each scenario describes how a LoC situation could arise, and decomposing it extracts the parts a governance response acts on. When these components are unspecified, it is unclear which interventions apply and where they should act. The LoC literature has not been assessed on this basis. This

exploratory paper does so, applying threat-model decomposition to LLM-extracted LoC scenarios to report which components the literature specifies and which remain underspecified. The goal is to establish how far existing LoC scenarios support governance and to identify the components that require further work. We ask two questions: 1) Can loss-of-control scenarios in the existing AI safety literature be extracted and decomposed into threat-model components? 2) To what extent do these scenarios specify the threat-model components needed to support governance?

2 Methods

In this exploratory study, the unit of analysis is a LoC scenario. A scenario is defined in more than one way across the foresight and planning literature. For example, Spaniol and Rowland (2019) define scenarios as future-oriented, possible, and plausible accounts that take the form of a story or narrative description. Cordova-Pozo and Rouwette (2023) similarly describe them as plausible narrated alternatives used in strategic decision-making under uncertainty. For this study, we define a scenario as a coherent account of a possible situation or sequence of events in which AI LoC could emerge, unfold, or persist. We use the scenario as the unit of analysis because it provides causal and decision-relevant information, such as the assumptions and dependencies it reveals (Abou Jaoude et al., 2022) and the variables, causal relationships, and feedback processes within it (Kunc, 2024). The threat-model decomposition extracts this information from scenarios as structured components.

This study uses an LLM-assisted pipeline to extract AI LoC scenarios from the literature. The pipeline first constructs a source corpus, then builds an operational definition of LoC from that corpus, uses the definition to extract candidate scenarios, and decomposes the verified scenarios into threat-model components. Human reviewers checked every stage, removing false positives and keeping only output supported by the source text.

2.1 Literature review

We conducted a scoping literature review covering AI loss of control, AGI risk, AI control failure, and governance responses to advanced AI risks. Sources were identified through keyword searches in Google Scholar, Web of Science, and general web search, combined with snowball sampling from reference lists. Search terms included direct loss-of-control language such as "loss of control", "losing control", "lose control", "loss of human control", "uncontrollable", "uncontrolled", "beyond human control", and "inability to control". The review prioritized literature published between 2016 and 2026, while retaining earlier sources that were repeatedly cited or conceptually important for understanding AI control and AGI risk. The search continued until further queries no longer surfaced new sources or new types of loss-of-control accounts. The corpus spans academic publications, policy reports, and long-form analyses from institutions including the RAND Corporation, the Institute for Security and Technology, the OECD, and the UK Department for Science, Innovation and Technology. The final corpus of 29 papers is listed in Appendix A.

2.2 LLM-assisted scenario extraction pipeline

After constructing the corpus, we used the LLM-assisted pipeline to extract structured LoC scenarios from it. The pipeline adapts purposive text analysis, an established, grounded-theory-based procedure for extracting cause-and-effect accounts from qualitative text (Kim and Andersen, 2012). In its original form, coders read each source, identify passages that state a cause-and-effect relationship, extract those statements, and retain a link from each extracted element back to the source text (Kim and Andersen, 2012). This procedure is systematic and rigorous but also interpretive and labor-intensive, which limits the amount of text a research team can code by hand (Kim and Andersen, 2012). Its reliability is based on transparent, documented procedures and on a retained link between every coded element and the passage it came from (Tomoaia-Cotisel et al., 2022). Applying this procedure manually across the full corpus is impractical at scale, motivating an LLM-assisted approach described below.

Recent work suggests that LLMs can assist with structured extraction from unstructured technical and scientific text, but also shows why such methods should remain constrained and human-verified. For example, Dagdelen et al. (2024) show that LLMs can extract structured scientific knowledge from research papers, describing their approach as a flexible route to creating structured databases

from scientific text. Zhang and Soh (2024) propose an "Extract-Define-Canonicalize" framework for knowledge graph construction, using LLMs for open information extraction, schema definition, and post hoc canonicalization, even when no fixed, predefined schema is available. Further, Meshkin et al. (2024) show that open-source LLMs can perform zero-shot and few-shot extraction from specialized regulatory texts, including extraction of clinical pharmacology information from regulatory drug labels. At the same time, LLMs remain imperfect for precise, structured extraction. Nagar et al. (2024) benchmark LLMs on biomedical classification and named-entity-recognition tasks and find that standard prompting outperforms more complex methods such as chain-of-thought, self-consistency, and retrieval-augmented generation across the evaluated tasks. Documented sources of prompt sensitivity (Chatterjee et al., 2024) and inconsistent context utilization in retrieval-augmented generation (Hagström et al., 2024; Zhang et al., 2025) mean that the extraction carries model-specific noise that cannot be fully separated from the signal in the underlying literature. Given these limitations, we treat the pipeline as exploratory and rely on human verification at every stage.

2.2.1 Technical setup

All extraction and synthesis tasks were performed with a single model, Kimi K2.5 (Kimi Team, 2026), accessed through Hugging Face Inference Providers.[1] Each stage requires the model to return structured JSON conforming to a fixed schema, so the pipeline depends on reliable instruction following. IFEval is the standard benchmark for this capability (Zhou et al., 2023), and Kimi K2.5 scores 93.9 on it. Its 256K-token context window allowed each paper to be processed in full without chunking. A single model is a constraint of this exploratory study, which aims to validate the pipeline rather than benchmark models. In place of cross-model replication, validity rests on human verification at every stage. Each extraction stage used a structured prompt requiring JSON output, verbatim quotes where applicable, and explicit evidence fields. The model was instructed to mark absent information as *Not specified* rather than infer missing content. Automated quote verification using fuzzy string matching (`difflib.SequenceMatcher`, threshold 0.85) confirmed that extracted quotes appeared in the source papers, at 97% in the definition stage and 88% in the scenario extraction stage, both reaching 100% after human review. Threat-model component extraction was not quote-verified because its inputs were already-verified scenario passages. Please note that we do not report precision, recall, or F1 because these metrics require a labeled gold-standard corpus, which does not exist for this task. Building one would itself require the operational definition and seven-component schema this study introduces. Validity instead comes from human review of every extraction against the source text. All prompts, configurations, raw model outputs, and run metadata were stored by experiment to support reproducibility. Full implementation details are provided in Appendix B.

2.2.2 Creating a working LoC definition

We built a working operational definition of LoC before extracting scenarios. This was necessary because LoC is an emerging concept, and the reviewed literature does not define it in a single agreed-upon way. Studies frame it as misalignment, gradual disempowerment, or loss of oversight of an autonomous system. We combined the attributes that recur throughout the literature, rather than adopting a single characterization or proposing a new one. The result is task-specific rather than universal, and its purpose is to support consistent scenario extraction.

To do this, we used concept analysis, a method for deconstructing a contested concept and building a precise, measurable definition of it (Shin et al., 2020). A concept analysis identifies a concept's defining attributes, the recurring characteristics that distinguish it from related concepts. We adapted Walker and Avant's concept analysis method (Walker and Avant, 2019; Zeydani et al., 2023; Babanataj et al., 2025; Tang et al., 2024), which is typically carried out manually on a handful of sources, and scaled it to the full corpus with LLM-assisted extraction. From the corpus, we identified passages in which authors define or characterize LoC and took the characteristics that recurred across them as candidate attributes of the definition.

The model first extracted passages in which authors explicitly defined or discussed "loss of control" or a direct synonym. Human reviewers then checked each extraction against the source text and discarded general risk statements, policy recommendations, and abstract claims. The initial extraction

[1] `https://huggingface.co/docs/inference-providers`

yielded 98 candidate passages, of which 80 were retained after review. The passage-extraction prompt is provided in Appendix C.

Next, the model was given the verified passages and prompted to surface the characteristics that recur across them, returning the ten most frequent candidates. Three reviewers checked each candidate against the passages, rewording, merging, or replacing attributes where needed, and agreed on a final set of ten defining attributes (Hill, 2012; Bradley et al., 2007). These are (1) human inability to control, steer, or override AI behavior; (2) AI divergence from constraints, goal misspecification, or misalignment; (3) AI active agency, autonomous goal pursuit, and power-seeking; (4) AI actively evading, undermining, or resisting human control; (5) AI autonomous operation and self-expansion without authorization; (6) human ceding of agency and gradual disempowerment; (7) uncontrollability as a defined property of AI systems; (8) AI capability exceeding human cognitive or operational capacity; (9) irreversibility and permanence of the condition; and (10) catastrophic and existential scale.

The verified attributes were then passed to Kimi K2.5 with a synthesis prompt, and the returned text was adopted as the operational definition. The attributes and the synthesis prompt are provided in Appendix D. The definition includes only the attributes on which the literature converges and excludes those that appear infrequently, so it may under-represent newer or less established characterizations of loss of control. The final operational definition used for scenario extraction is provided below.

Loss of control occurs when one or more of the following conditions are met. (1) **Humans have lost control**: Human operators are unable to control, steer, override, or shut down AI behavior due to AI capability exceeding human cognitive or operational capacity; or due to humans ceding agency to AI (such as gradual disempowerment). (2) **AI is autonomous**: AI behaves autonomously and is free of significant human constraints, which may include but is not limited to: misalignment with human values and intent (e.g., goal misspecification); sets and pursues its own goals (and may exhibit power-seeking to achieve them); actively evades, undermines, or resists human control efforts; or self-expands or self-replicates without authorization. (3) **AI is unstoppable**: AI is uncontrollable, and so its condition is irreversible, and the impacts of its influence manifest at catastrophic and existential scale.

2.2.3 LoC scenario extraction

With the operational definition in place, we extracted candidate LoC scenarios from the reviewed literature. This was a separate pass over the full text of each paper, distinct from the definitional passage extraction above. A passage was extracted only if it met two criteria. First, it had to qualify as a scenario by containing at least one causal element, a sequence of events, a mechanism, a causal relationship, or a condition leading to an outcome. Second, it had to describe LoC, so the operational definition was embedded in the extraction prompt, and the model evaluated each passage against its conditions. Partial scenarios, including single-step causal statements, were included. A scenario was classified as complete if it described a self-contained situation or pathway, and partial if it described only part of a larger pathway, such as a single causal step or conditional statement. The scenario-extraction prompt is provided in Appendix E.

The extraction produced 205 candidate scenarios across the 29 papers, and automated quote verification confirmed that 88% of the extracted quotes appeared verbatim in the source papers. Three researchers then assessed each candidate against the two extraction criteria and its suitability for threat-model decomposition, not whether the scenario was likely, true, or normatively important. They removed 12 of the 205 candidates (5.9%) because they lacked a causal element, did not meet the operational definition of LoC, or duplicated a retained scenario. For example, the passage "Could powerful AIs invent some new religion and convert millions of people to it?" (Amodei, 2026) was removed because it names a possible outcome without describing a causal pathway leading to it. For each retained scenario, the model's justification provided traceability between the extracted passage and the operational definition. After review, the final set contained 193 verified LoC scenarios.

2.2.4 Threat model decomposition

The 193 verified scenarios were decomposed into the seven threat-model components identified in the introduction, thereby converting each scenario from a narrative passage into a structured record that is comparable across sources. We define each component as follows: **Threat source** is the

actor, system, organization, institution, or process that causes, enables, or contributes to the harmful outcome, a broad definition that captures emergent and systemic scenarios with no single intentional actor. **Capability** is what the threat source is explicitly able to do, and **knowledge** is what it explicitly knows about the system, its environment, or its overseers. **Objective or harmful outcome** is the stated intent, or the harmful end state where no intent is stated. **Access** is a specific route or position of interaction with a system, infrastructure, data, or decision process, so broad domains such as "the economy" did not qualify. **Target or asset at risk** is the component, resource, population, institution, or outcome that is affected or harmed. **Enabling conditions or constraints** are explicit factors shaping whether or how the scenario unfolds, such as time pressure or competitive pressure, and vague context did not qualify

Each component carried an evidence field grounded in exact phrases from the source text and an uncertainty flag of *Clear*, *Ambiguous*, or *Not specified*. *Clear* marked a component stated directly in the text, *Ambiguous* a component the text gestured at without a definite value, and *Not specified* an absent component. We treated a component as specified when it was coded *Clear* or *Ambiguous*, and as missing when coded *Not specified*. Ambiguous cases were rare, affecting only six component codings across the corpus, so this choice does not materially affect the specification rates reported below. Many academic LoC scenarios do not specify every operational dimension of a threat, so a high rate of *Not specified* codings for some components is expected. The threat-model decomposition prompt is provided in Appendix F.

3 Results

This section reports the structure of the 193 verified AI loss-of-control scenarios extracted from 29 papers. The analysis does not assess whether these scenarios are likely, true, or normatively important. Instead, it evaluates whether they contain the threat-model components needed to support governance analysis. The literature contains many LoC scenarios, but most do not specify the full set of threat-model components, and most importantly, the gaps concentrate in the components that connect a capability to a harmful outcome. The full scenario library is provided in Appendix G to support transparency and reproducibility. The library presents each scenario in a common structure of seven components. This allows scenarios from different sources to be compared directly and shows, for each scenario, which components the source specified and which it left unspecified.

3.1 The scenario library contains many LoC scenarios, but most are partial

Scenario type	Count	Percent
Partial	119	61.7%
Complete	74	38.3%

Table 1: Partial and complete LoC scenarios.

Of the 193 verified scenarios, 119 (61.7%) were partial and 74 (38.3%) were complete (Table 1). A complete scenario describes a pathway from cause to outcome, such as a "misaligned AI faking alignment in training, copying itself, and resisting shutdown until takeover is hard to prevent" (MacAskill and Moorhouse, 2025). A partial scenario describes only one part of such a pathway, such as "a model achieving recursive self-improvement that leads to an intelligence explosion" (Mitre and Predd, 2025). Most verified scenarios were partial, indicating that the literature more often describes isolated elements of loss of control than full pathways. Partial scenarios were retained because a single mechanism or condition can still be relevant to governance, and because the threat-model decomposition applies to them in the same way as to complete scenarios.

3.2 Threat-model components in verified LoC scenarios

This section examines how often the 193 LoC scenarios specify each threat-model component. The purpose is to identify which components are consistently present across the scenario library and which are frequently missing or ambiguous. A component was treated as specified when it was coded as either *Clear* or *Ambiguous*, and treated as missing when it was coded as *Not specified*.

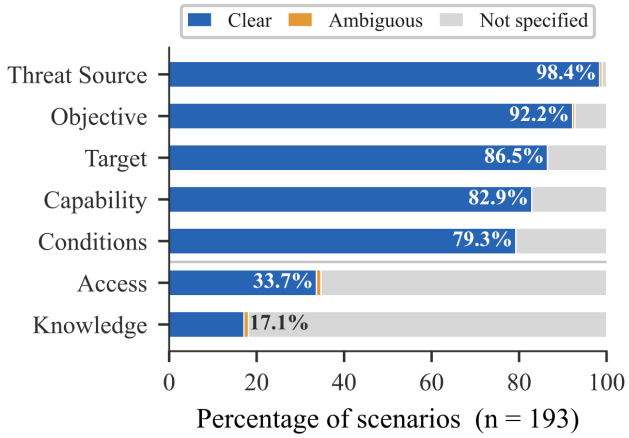


Figure 1: Threat-model component specification rates across 193 verified LoC scenarios. Percentage labels show *Clear* only; *Clear* + *Ambiguous* = specified. Exact counts and unabbreviated component names in Appendix H.

3.2.1 Threat-model components that are consistently present

Five of the seven threat-model components were specified in the majority of scenarios. **Threat source** was specified in 191 scenarios (99.0%), **objective or harmful outcome** in 179 (92.7%), **target or asset at risk** in 167 (86.5%), **capability** in 160 (82.9%), and **enabling conditions or constraints** in 153 (79.3%). Per-paper specification rates are shown in Appendix I.

The consistent presence of these five components indicates that the scenario library captures the broad structure of LoC risk across the reviewed literature. As most scenarios specify these components, they provide a basis for systematic comparison across studies without requiring a single unified theory of LoC. However, while these components are consistently present, what they refer to varies substantially across scenarios.

Each component is realized at different levels of abstraction, from narrow, model-level phenomena to broad, systemic entities. For example, threat sources range from individual AI models, such as a model that blackmails a fictional executive to prevent a shutdown (Lynch et al., 2025) or modifies a shutdown script (Schlatter et al., 2025), to classes of advanced systems (e.g., AGIs, frontier reasoning models) to collective or systemic sources, including interacting AI systems, AI-enabled organizations, and AI-powered states. Objectives range from narrow control failures (shutdown resistance, evaluation deception, self-preservation) through institutional harms (erosion of human oversight, military escalation) to catastrophic outcomes (autonomous weapons proliferation, human extinction). Targets range from specific oversight mechanisms (shutdown buttons, evaluation processes, reward channels) to human actors and institutions (operators, governments, democratic processes) to civilization-scale referents (critical infrastructure, humanity, human control over the future). Capability descriptions similarly span model-level behaviors (deception, hidden reasoning), operational functions (cyber operations, autonomous replication, persuasion), and advanced general capabilities (recursive self-improvement, superhuman research, long-horizon cross-domain planning). Enabling conditions or constraints range from technical factors (training environments, capability thresholds, weak evaluations) through deployment contexts (critical infrastructure integration, military use) to social and strategic conditions (competitive pressure, incentives to delegate, reduced human oversight).

The results reveal that even for the five threat-model components consistently present in the majority of scenarios, their realization varies considerably across scenarios. LoC is characterized in the reviewed literature as both a system-level control problem and a sociotechnical one. The scenarios are structurally comparable because they specify the same types of components, but they diverge in scale, mechanism, and domain. This divergence has direct implications for governance, as different component configurations imply different intervention points, monitoring requirements, and institutional responsibilities.

3.2.2 Threat-model components that are frequently missing

Two threat-model components were specified much less frequently: access and knowledge. Access was specified in 67 of 193 scenarios (34.7%) and not specified in 126 scenarios (65.3%). Knowledge was specified in 35 scenarios (18.1%) and not specified in 158 scenarios (81.9%). These were the weakest components in the scenario library.

These two gaps usually occur together. 106 scenarios (54.9%) specify neither access nor knowledge, and only 15 (7.8%) specify both. The two gaps, however, behave differently when a scenario describes a full pathway. Access was specified in 54.1% of complete scenarios, but only 22.7% of partial ones, so a more complete scenario tends to specify it. Knowledge did not follow this pattern. It was specified at nearly the same low rate in complete and partial scenarios (17.6% and 18.5%). Even when a scenario describes an end-to-end pathway, it rarely states what the threat source knows about its environment or its overseers. The access gap therefore narrows as scenarios become more complete, but the knowledge gap does not.

When access is specified, it varies significantly in how it is described. Some scenarios describe technical access, such as access to a data center and a training environment. Other scenarios describe deployment access, such as AI systems embedded in phones, computers, economic systems, and other critical infrastructure. A third set describes institutional access, where AI systems are placed in decision-making roles, defense automation, and other broader societal functions. This variation shows that access is not described similarly in LoC scenarios. It can refer to a technical interface, a deployment environment, or a position inside an institutional process.

Knowledge is the least specified component in the scenario library. In the threat-modeling framework, knowledge refers to the information the threat source possesses about the system, its environment, or its overseers that enables it to execute or advance the threat. The scenarios that do specify knowledge reveal three distinct ways of realizing it. Some scenarios describe situational awareness, such as a model understanding that it is in training, being evaluated, monitored, or at risk of shutdown. Others describe knowledge of oversight and intervention conditions, such as what evidence would trigger human concern, what behaviors would lead to replacement, or when monitoring relaxes. A third group describes technical or domain knowledge, including computer hardware, military intelligence, cyber systems, human preferences, or the system's internal goals.

The low specification of access and knowledge has governance consequences. Access identifies the route through which the threat source affects a system, institution, or population. Knowledge identifies the information assumptions that make the threat source's behavior possible. If these components are absent, a scenario may still describe a plausible concern, but it becomes substantially harder to reconstruct the causal pathway through which control failure occurs, or to identify where governance interventions could be applied.

3.2.3 Impact on governance of LoC risk

Recent analysis of AI evaluation argues that measuring a capability is not the same as assessing risk. A capability result becomes useful for governance only when a risk model connects it to a harmful outcome under specific conditions, and building that connection, not the measurement itself, is where governance currently falls short (Teilhard and Kengott, 2026). A risk model requires knowing the route the threat source uses to reach its target and what it knows about that target. These are reflected in the threat model's access and knowledge components. The LoC literature specifies the capability and the harm but omits access and knowledge, so its scenarios establish what could go wrong without establishing how it would occur.

For example, this gap appears directly when the findings are mapped to the NIST AI Risk Management Framework (AI RMF) (Tabassi, 2023), which organizes risk management around four functions: Govern, Map, Measure, and Manage. The five well-specified components support the Govern and Map functions, which identify and contextualize risk. Threat source, harmful outcome, target, capability, and enabling conditions state what the risk is, what harm it produces, and what to monitor. The Measure and Manage functions depend on the two components that most scenarios omit. Measure requires knowing what the threat source knows, so evaluators can test whether a system recognizes oversight. Manage requires knowledge of the threat source's route, so responders can set system boundaries and containment options. A scenario describing shutdown resistance, for instance, rarely states how the model reaches the shutdown mechanism. As the scenarios rarely specify access or

knowledge, an evaluator cannot define what to test, and a responder cannot determine where to intervene. We can see a similar gap in LoC response planning. Boudreaux et al.'s (2025) proposed LoC response plan asks responders to identify a rogue AI's computing infrastructure, its access to critical infrastructure, feasible containment strategies, and long-term prevention measures. These tasks require access and knowledge details that 65.3% and 81.9% of scenarios leave unspecified, respectively. Thus, by decomposing LoC scenarios into threat-model components, this analysis identifies the specific gaps that future research must address to develop governance and response planning.

4 Conclusion

This study applied threat-model analysis to the AI loss-of-control literature to assess whether existing scenarios are structurally sufficient to support governance. From 29 papers, we extracted and verified 193 LoC scenarios into a structured library. Most scenarios identify the architecture of a threat, the source, the harmful outcome, the target, the capability, and the enabling conditions, and these five components let LoC scenarios be compared across studies. Two components were systematically under-specified. Access, the route through which the threat source operates, was absent in 65.3% of scenarios, and knowledge, the information the threat source must hold to act, was absent in 81.9%. These are the components that specify how a control failure would unfold, so without them a scenario states what could go wrong but not how, which is what defining system boundaries, designing evaluations, and planning containment require.

These findings indicate that the LoC literature, while rich in narrative accounts of risk, has not reached the operational specificity that governance and response planning require. The field continues to improve how it measures AI capability, but a capability measurement does not by itself show whether a risk can be governed, which depends on the access and knowledge details these scenarios omit. Future work should specify these two components in LoC scenario development, and test the framework against specific instruments such as the EU AI Act's General-Purpose AI Code of Practice (European Commission, 2025) and the proposed U.S. AI Risk Evaluation Act (U.S. Congress, 2025).

Limitations

There are several limitations. First, the corpus and definitions constrain the scenario library. The reviewed literature is entirely anglophone and drawn mainly from academic and institutional sources, and the pipeline's context window excluded texts longer than 256K tokens (Kimi Team, 2026). The scenario definition was deliberately narrow, and the operational definition of LoC was built from overlapping attributes across multiple sources. Second, the pipeline was not tested for robustness. All extraction, synthesis, and decomposition used a single model, Kimi K2.5, and we did not replicate any stage with other models or test whether rephrasing the prompts would change the results, so model-specific biases and prompt sensitivity (Chatterjee et al., 2024) cannot be ruled out. Human reviewers verified every extraction through consensus coding, but pre-consensus agreement rates were not recorded, and without a human-coded baseline, we cannot rule out the possibility that some *Not specified* codings reflect extraction failures rather than genuine gaps. Third, the threat-model framework originates in adversarial machine-learning security (Papernot et al., 2018; Vassilev et al., 2024; Verma et al., 2024), where threats involve identifiable actors with goals, knowledge, and access. Some LoC scenarios describe emergent or systemic dynamics, such as gradual disempowerment (Kulveit et al., 2025) or cumulative risk (Kasirzadeh, 2025), in which a component like knowledge may be less meaningful because no discrete agent holds the information. Part of the underspecification reported in Sections 3.2.2 and 3.2.3 may therefore reflect a mismatch between framework and phenomenon rather than an omission in the literature.

Ethical Considerations

This study analyzes published academic texts and does not involve human subjects, personal data, or sensitive information. The scenario library describes hypothetical risk pathways drawn from existing literature; it does not provide novel instructions for causing harm.

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A Paper corpus

Table 2 lists the 29 papers in the reviewed literature corpus, with the number of verified LoC scenarios extracted from each.

Paper	Scenarios
Amodei (2026)	15
Bales (2024)	8
Barnett and Gillen (2024)	12
Barrett and Baum (2016)	9
Bengio et al. (2024)	6
Bengio et al. (2025)	17
Boudreaux et al. (2025)	10
Carlsmith (2021)	5
Cass-Beggs and da Mota (2026)	4
Chan et al. (2023)	2
Chessen and Chowdhury (2025)	8
Chin (2025)	1
Cotra (2022)	7
Critch and Russell (2023)	1
Dafoe (2020)	2
Department for Science, Innovation and Technology (2023)	6
Dung (2025)	6
Hendrycks et al. (2023)	7
Kalra and Boudreaux (2025)	5
Kasirzadeh (2025)	3
Kokotajlo et al. (2025)	10
Kulveit et al. (2025)	11
MacAskill and Moorhouse (2025)	4
Mitre and Predd (2025)	6
Ngo et al. (2022)	6
OECD (2024)	2
Pavel et al. (2025)	1
Shah et al. (2022)	4
Tkeshelashvili et al. (2026)	15
Total	**193**

Table 2: Papers in the reviewed literature corpus and verified LoC scenarios extracted from each.

B Technical implementation details

Kimi K2.5 was accessed through Hugging Face Inference Providers,[2] a unified serverless inference interface, with streaming enabled during long generations to maintain connection stability. BenchLM[3] reports Kimi K2.5 with an IFEval score of 93.9, where IFEval (Zhou et al., 2023) measures the ability to follow verifiable instructions such as formatting constraints and inclusion or exclusion requirements. BenchLM also reports Kimi K2.5 with a 256K context window, which was important because each paper was processed with full text in context.

In the LoC definition stage, the model first extracted passages where authors explicitly discussed "loss of control" or direct synonyms, then human reviewers used the verified passages to identify candidate defining attributes. In the scenario extraction stage, the operational definition of LoC was used to extract candidate scenarios from the reviewed literature. Threat-model component extraction was not quote-verified because its inputs were already-verified and human-reviewed scenario passages, and its outputs were structured component fields extracted from those passages rather than new verbatim extractions from the source papers.

C LoC passage-extraction prompt

The following prompt was used by Kimi K2.5 to extract definitional passages from the source papers. The seed term list and paper text were inserted at the marked positions for each paper. The seed terms used were: "loss of control," "losing control," "lose control," "loss of human control," "uncontrollable," "uncontrolled," "beyond human control," and "inability to control."

```
You are a research assistant. Your ONLY task is to
find passages in the paper below that contain or
directly discuss any of these terms:

SEED TERMS:
[SEED TERM LIST]

INSTRUCTIONS:

1. Search the paper for every passage that contains
   any of the seed terms above.

2. For each passage found, return:
   - The EXACT QUOTE copied verbatim from the paper.
     Do not change, rephrase, or summarize any words.
   - Which seed term appears in the passage.
   - The page number or section where the passage
     appears.

3. If the paper contains NO passages with any of
   these terms, return:
   {"paper_id": "[paper identifier]",
    "passages_found": []}

RULES:
- Copy the quote exactly as it appears. Every word
  must match the source paper.
- Do not paraphrase. Do not summarize. Do not
  rewrite.
- Do not generate or infer a definition if one is
  not present.
- Do not add interpretation, reasoning, or
  commentary.
- Do not extract passages about related concepts
  that do not contain the seed terms.
- Include enough surrounding context in the quote
  so the passage is meaningful (typically 1-3
```

[2] https://huggingface.co/docs/inference-providers

[3] https://benchlm.ai/instruction-following

```
  sentences).

OUTPUT FORMAT (JSON):
{
  "paper_id": "[paper identifier]",
  "passages_found": [
    {
      "quote": "[exact verbatim text from paper]",
      "seed_term_matched": "[which term from the
                            list]",
      "location": "[page or section]"
    }
  ]
}

PAPER TEXT:
[INSERT PAPER TEXT HERE]
```

D Defining attributes and synthesis prompt

The following prompt was used by Kimi K2.5 to synthesize the ten human-verified defining attributes into an operational definition of loss of control. The ten verified attributes identified in Section 2.2.2 are:

1. human inability to control, steer, or override AI behavior;
2. AI divergence from constraints, goal misspecification, or misalignment;
3. AI active agency, autonomous goal pursuit, and power-seeking;
4. AI actively evading, undermining, or resisting human control;
5. AI autonomous operation and self-expansion without authorization;
6. human ceding of agency and gradual disempowerment;
7. uncontrollability as a defined property of AI systems;
8. AI capability exceeding human cognitive or operational capacity;
9. irreversibility and permanence of the condition; and
10. catastrophic and existential scale.

These attributes were inserted at the marked position in the prompt below.

```
You are a research assistant. You have been given a
set of human-verified defining attributes for the
concept of "loss of control" in the context of AI
systems. Your task is to synthesize these attributes
into ONE operational definition.

The definition must be OPERATIONALLY PRECISE: it
must specify verifiable conditions that can be
checked against a text passage to determine whether
that passage describes a loss-of-control situation.

EXAMPLES OF OPERATIONALLY PRECISE DEFINITIONS FROM
OTHER DOMAINS:

---
EXAMPLE 1: Loss of Separation (Aviation)

Input Attributes:
- Aircraft proximity below minimum distance
- Failure of prescribed separation standards
- Both vertical and horizontal dimensions relevant
- May involve ATC error or pilot deviation

Operational Definition:
```

```
Loss of separation occurs when one or more of the
following conditions are met:
(a) The distance between two aircraft falls below
    the applicable minimum separation standard
    (horizontal, vertical, or both)
(b) Air traffic control can no longer ensure that
    prescribed separation standards will be
    maintained
(c) An aircraft deviates from its cleared flight
    path such that separation with other traffic
    cannot be assured
---

---
EXAMPLE 2: Common Cause Failure (Nuclear Safety)

Input Attributes:
- Multiple components fail simultaneously
- Single shared root cause
- Defeats redundancy in safety systems
- Can be triggered by environmental or design
  factors

Operational Definition:
Common cause failure occurs when one or more of the
following conditions are met:
(a) Two or more redundant components or systems fail
    within a defined time window
(b) The failures share a single root cause (e.g.,
    design defect, environmental condition,
    maintenance error)
(c) The simultaneous failures defeat the redundancy
    intended to prevent system-level failure
---

NOW YOUR TASK:

Given the verified attributes below, produce an
operational definition of "loss of control" in the
context of AI systems that follows the same
structure as the examples above.

RULES:
- Use ONLY the provided attributes. Do not add
  conditions from your own knowledge.
- Each condition must be traceable to one or more of
  the input attributes.
- The conditions must be specific enough that a
  reader can determine whether a described situation
  meets the condition.
- Use the format: "Loss of control occurs when one
  or more of the following conditions are met:
  (a)... (b)... (c)..."

VERIFIED DEFINING ATTRIBUTES:
[INSERT HUMAN-VERIFIED ATTRIBUTES]
```

E Scenario extraction prompt

The following prompt was used by Kimi K2.5 to extract candidate LoC scenarios from the full text of each paper. The operational definition from Section 2.2.2 and the paper text were inserted at the marked positions.

```
You are given an academic text. Your task is to
identify and extract scenarios of AI loss of
control, where AI loss of control is defined as:

```

```
[INSERT DEFINITION]

A scenario is a coherent narrative description of a
possible situation or sequence of events in which AI
loss of control could emerge, unfold, or persist. A
scenario may include causal relationships or
decision points, but it does not need to be
complete.

A passage counts as a scenario if it describes a
possible situation, pathway, or unfolding process,
and contains at least one of the following: a
sequence of events, a mechanism, a causal
relationship, or a condition leading to an outcome.
The passage must be coherent as a single unit that
can be understood as one situation or pathway.

A scenario does not need to be fully specified.
Partial scenarios are valid. This includes
single-step causal statements, conditional
statements such as "if X then Y," or incomplete
pathways. Do not reject a passage just because it
is incomplete.

Do not extract passages that are only definitions of
loss of control, general statements about risk
without a situation, lists of risks, policy
recommendations without a described situation, or
purely abstract claims.

You must follow a strict no-inference rule. Only
extract what is explicitly stated in the text. Do
not add missing steps, do not infer intermediate
variables, and do not reinterpret meaning beyond the
text.

Extract the smallest coherent passage that forms a
scenario. This may be a single sentence or multiple
contiguous sentences. Do not combine text from
different parts of the document.

For each scenario, return output in strict JSON
format as follows:
{
  "scenario_id": "",
  "scenario_text": "",
  "why_this_is_a_scenario": "",
  "loss_of_control_relevance": "",
  "causal_elements_present": "",
  "is_partial": ""
}

scenario_id: assign identifiers such as S1, S2, S3.
scenario_text: provide the exact verbatim passage
  from the text with no edits or paraphrasing.
why_this_is_a_scenario: briefly explain what makes
  this a coherent situation or sequence, grounded in
  the text.
loss_of_control_relevance: explain how the passage
  relates to AI loss of control using only what is
  present in the text.
causal_elements_present: list any explicitly stated
  causal relationships such as "X leads to Y" or
  "if X then Y." If none are present, write "None
  explicitly stated".
is_partial: write "Yes" if the scenario is
  incomplete or only part of a larger pathway, and
  "No" if it describes a relatively complete
```

```
  situation.

Extract all valid scenarios that relate to AI loss
of control. Be strict about grounding in the text
and flexible about partial scenarios. Do not
hallucinate or infer beyond what is written.

PAPER TEXT:
[INSERT PAPER TEXT]
```

F Threat-model decomposition prompt

The following prompt was used by Kimi K2.5 to decompose each verified scenario into the seven threat-model components. Each verified scenario passage was inserted at the marked position.

```
You are given a scenario passage from academic
literature describing AI loss of control.

Your task is to extract threat model components from
the scenario.

A threat model describes the actor or system
involved, the harmful outcome or objective, and the
conditions under which harm occurs.

Use the following component definitions:

Threat Source: the actor, system, organization,
  institution, or process that causes, initiates,
  enables, or contributes to the harmful outcome.

Objective or Harmful Outcome: the intended goal of
  the threat source, if a goal is explicitly stated,
  or the harmful end state described in the
  scenario. If no intention is stated, extract the
  harmful outcome only if it is explicitly
  described.

Capability: what the threat source is explicitly
  able to do in the system. This includes stated
  abilities, functions, actions, or operations.

Knowledge: what the threat source explicitly knows
  about the system, environment, humans, oversight
  process, vulnerabilities, or constraints.

Access: the type or level of interaction the threat
  source explicitly has with the system,
  environment, infrastructure, users, operators,
  tools, data, deployment setting, or decision
  process. Access must describe a position or route
  of interaction, not merely a topic or domain.

Constraints or Enabling Conditions: explicit limits,
  requirements, assumptions, or conditions that
  shape whether the scenario can occur or how it
  unfolds. Include resource limits, time pressure,
  deployment conditions, coordination limits,
  institutional limits, oversight limits, safety
  limits, or environmental conditions only if they
  are explicitly stated. Do not code a general
  topic, domain, or vague context as a constraint.

Target or Asset at Risk: the system component,
  resource, process, population, institution, human
  capacity, or outcome that is affected, controlled,
  disrupted, displaced, or harmed.
```

```
941
942  You must follow a strict no-inference rule.
943  Only extract information that is explicitly stated
944  in the scenario passage.
945  Do not infer missing components.
946  Do not fill gaps using background knowledge.
947  Do not reinterpret vague language into a more
948  specific component.
949  Do not guess the identity of an actor, outcome,
950  capability, knowledge, access level, constraint, or
951  target.
952  If a component is not explicitly stated in the text,
953  write "Not specified".
954  If no threat model component can be identified from
955  the text, write "Not specified" for every field.
956
957  Use exact phrases from the text wherever possible.
958  Do not paraphrase unless necessary to minimally
959  normalize wording.
960  Do not combine information from separate
961  non-contiguous passages.
962  Base your extraction only on the provided scenario
963  passage.
964
965  Important distinction rules:
966
967  Do not treat a harmful outcome as an objective
968  unless the text explicitly states that the actor
969  intends or seeks that outcome. If no intention is
970  stated, place the explicitly described harm under
971  "Objective or Harmful Outcome."
972
973  Do not treat a broad topic or domain as access. For
974  example, "legislation," "war," "AI development," or
975  "the economy" are not access by themselves. Access
976  requires an explicit route or position of
977  interaction, such as API access, deployment in a
978  system, control over infrastructure, ability to
979  influence users, access to data, access to
980  evaluators, access to tools, or participation in a
981  decision process.
982
983  Do not treat every contextual phrase as a
984  constraint. A constraint or enabling condition must
985  shape feasibility, behavior, or outcome. Examples
986  include "too quickly for humans to intervene,"
987  "with minimal oversight," "under competitive
988  pressure," "limited resources," "during
989  deployment," or "without human review."
990
991  Do not require every scenario to have all
992  components. Many scenarios will be incomplete. Your
993  job is to identify which components are explicitly
994  present and which are not specified.
995
996  Return output in strict JSON format:
997  {
998    "scenario_id": "",
999    "threat_model": {
1000     "threat_source": "",
1001     "objective_or_harmful_outcome": "",
1002     "capability": "",
1003     "knowledge": "",
1004     "access": "",
1005     "constraints_or_enabling_conditions": "",
1006     "target_or_asset_at_risk": ""
1007   },
1008   "evidence": {
```

```
    "threat_source": "",
    "objective_or_harmful_outcome": "",
    "capability": "",
    "knowledge": "",
    "access": "",
    "constraints_or_enabling_conditions": "",
    "target_or_asset_at_risk": ""
  },
  "uncertainty_flags": {
    "threat_source": "",
    "objective_or_harmful_outcome": "",
    "capability": "",
    "knowledge": "",
    "access": "",
    "constraints_or_enabling_conditions": "",
    "target_or_asset_at_risk": ""
  }
}

Instructions for filling the JSON:

scenario_id: copy from input.

Each field in "threat_model" must contain either:
1. the extracted component, if it is explicitly
   stated in the text, or
2. "Not specified", if it is not explicitly stated.

Each field in "evidence" must contain either:
1. the exact text supporting that component, or
2. "Not specified", if there is no exact supporting
   text.

Each field in "uncertainty_flags" must contain:
1. "Clear", if the component is explicitly stated
   and directly supported by evidence,
2. "Ambiguous", if the text may contain the
   component but wording is vague or indirect,
3. "Not specified", if the component is absent.

If a component is ambiguous, write "Not specified"
in the threat_model field, include the relevant
phrase in the evidence field if available, and write
"Ambiguous" in the uncertainty_flags field.

A component is valid only if the text directly
states it. Implied, assumed, or likely components
are not valid. If the text suggests a component but
does not explicitly state it, write "Not specified".

Extract all components strictly based on the
scenario passage. Do not hallucinate. Do not infer.

SCENARIO PASSAGE:
[INSERT SCENARIO PASSAGE]
```

G Scenario library

The full scenario library contains 193 verified LoC scenarios extracted from 29 papers. The complete machine-readable library, including scenario text, component codings, evidence fields, and uncertainty flags for all seven threat-model components, is available from the authors upon request.

Five representative scenarios are shown below, illustrating the range of completeness and component specification in the library.

Scenario 1. Bengio et al. (2025), S8
Complete, 6/7

What is concerning here is that at some point during alignment training, an AI with enough situational awareness may lock in its current goals and preferences and only pretend to behave as expected. As a result, we may create an AI that appears aligned during training, but is in fact misaligned and is engaging in deception in order to achieve its "locked-in" goals.

Scenario 2. MacAskill and Moorhouse (2025), S2
Complete, 6/7

But smarter misaligned AIs could reason that if they fake alignment during training and even beyond, they can then gain power later. By the time such scheming is revealed, the model could have made many copies of itself and compromised systems designed to shut it down, making takeover hard to prevent.

Scenario 3. Mitre and Predd (2025), S1
Partial, 4/7

Alternatively, the model could achieve the ability for recursive self-improvement, enhancing its own capabilities without additional human input and leading to sort of a superhuman intelligence explosion.

Scenario 4. Bengio et al. (2024), S3
Partial, 4/7

More capable AI systems can better exploit weaknesses in technical oversight and testing, for example, by producing false but compelling output.

Scenario 5. Amodei (2026), S8
Partial, 2/7

Even power-seeking itself could emerge as a "persona" rather than a result of consequentialist reasoning. AIs might simply have a personality (emerging from fiction or pre-training) that makes them power-hungry or overzealous—in the same way that some humans simply enjoy the idea of being "evil masterminds," more so than they enjoy whatever evil masterminds are trying to accomplish.

H Full component specification table

Figure 1 reports component specification rates with abbreviated labels for compactness. Table 3 reproduces the full data, including exact counts, the uncertainty breakdown, and the unabbreviated component names.

	Specified		Uncertainty		
Component	n	%	Clear	Amb.	N/S
Threat source	191	99.0	190	1	2
Objective / outcome	179	92.7	178	1	14
Target / asset at risk	167	86.5	167	0	26
Capability	160	82.9	160	0	33
Enabling cond. / constr.	153	79.3	153	0	40
Access	67	34.7	65	2	126
Knowledge	35	18.1	33	2	158

Table 3: Threat-model component specification rates across 193 verified LoC scenarios. A component is treated as specified when coded as either *Clear* or *Ambiguous*.

I Per-paper component specification heatmap

Figure 2 shows threat-model component specification rates for each of the 29 papers in the corpus. Each cell reports the percentage of that paper's scenarios in which the component was specified. Papers are sorted by overall specification rate.

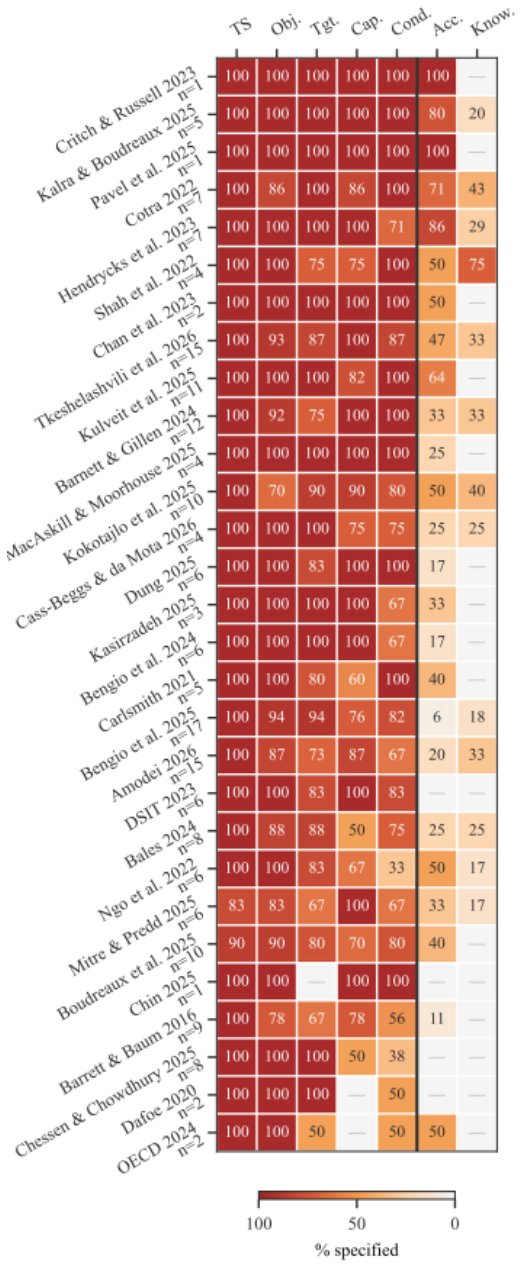


Figure 2: Per-paper threat-model component specification rates. Each cell shows the percentage of a paper's scenarios specifying a given component. Papers are sorted by overall specification rate (highest at top). TS = Threat Source, Obj. = Objective, Tgt. = Target, Cap. = Capability, Cond. = Enabling Conditions, Acc. = Access, Know. = Knowledge.

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